



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Department of Computer Science and Engineering

Course Coordinator	Dr. Giridhar Urkude		
Academic Year	2025-26	Semester	V
Course Title	Compiler Construction Lab		
Course Code	T7478/ 0705210507	Software/Platform	Lex, Yacc
Course Credit	4		

Course Objectives:

Course Objective No.	Course Objective Statements
1	To outline the details of LEX for identifying tokens using Regular Expressions.
2	To understand and implement YACC for grammars in languages.
3	To implement the semantic analyzer using rules in LEX and YACC programs.
4	To model an intermediate code (IC) generator for arithmetic expressions using YACC.

Course Outcomes:

Course Outcome No.	BT Level	Course Outcome Statements
		Students will be able to:
CO-1	BL 6	Design a lexical analyzer and parser for a given programming language syntax.
CO-2	BL 2	Interpret compiler construction tools like LEX and YACC.
CO-3	BL 6	Implement compiler components such as lexical analyzers and parsers using LEX and YACC.
CO-4	BL 6	Illustrate the ability to generate intermediate code for arithmetic expressions.



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Course Articulation Matrix

	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO-1	PSO-2
CO-1	3	3	3	2	2	–	–	1	1	–	2	3	1
CO-2	2	1	–	–	3	–	–	–	–	–	2	2	–
CO-3	3	2	3	2	3	–	–	1	1	1	2	3	1
CO-4	3	2	3	2	2	–	–	–	1	–	2	3	–
CO-5	3	3	3	2	2	–	–	1	1	–	2	3	1
Average	3	3	3	2	2	–	–	1	1	–	2	3	1

List of Practical's/Experiments:

S. No	CO(s) Mapped	Name of the Experiment
1	CO1	Introduction to LEX tool, metadata and patterns.
2	CO1	Count the number of comments, keywords, identifiers, words, lines and spaces from input file.
3	CO1	Count number of words starting with “A”.
4	CO2	Introduction to YACC tool, and format to write YACC code.
5	CO2	Conversation of lowercase to uppercase and vice versa.
6	CO2	Conversation of decimal to hexadecimal number in a file.
7	CO3	Test lines ending with “COM”.
8	CO3	Postfix Expression Evaluation.
9	CO4	Desk calculator with error recovery.
10	CO4	Parser for “FOR” loop statements



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S. No	CO(s) Mapped	Value Added Experiment
1	CO4	Intermediate code generator (ICG) for Arithmetic expression.

GitHub Link (If Applicable):

NA

Course Faculty Members:

1. **Dr. Giridhar Urkude**
2. Dr. Priya Dasarwar
3. Dr. Snehlata Wankhade
4. Prof. Rohit Pawar

Dr. Sagarkumar S. Badhiye
Head of the Department

Dr. Sandeep Kumar
Deputy Director

Dr. Nitin Rakesh
Director

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